

In Class B push – pull amplifier both transistor are biased at

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a. linear operation ☐

b. maximum power ☒

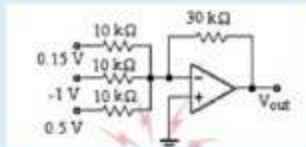
c. saturation ☐

d. cutoff ☐

?Which is greater, the breakdown voltage or the barrier potential .

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- a. Breakdown voltage is always much less than the barrier potential ☐
- b. Breakdown voltage is always equal the barrier potential ☒
- c. Breakdown voltage is always much greater than the barrier potential ☐
- d. none ☐



A voltmeter placed from the inverting input to ground would read

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a. 2.775 v ☐

b. -2.775 v ☒

c. 0 v ☐

d. -0.925 v ☐

: In JFET, the channel is between

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a. Gate and Drain ☒

b. Has no channel ☐

c. Source and Drain ☐

d. Gate and Source ☐

?What symptoms does an open R_C produce

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- a. If R_C is open $V_{CC} = V_{CE}$ ☐
- b. If R_C is open, V_C is about 0.7 V less than V_B ☒
- c. If R_C is open $V_{CC} = 0$ ☐
- d. If R_C is open $V_{CE} = 0$ ☐

For a class A CE amplifier, the power dissipation P_{DQ} is

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- a. is zero with no input signal ☐
- b. is maximum when there is no input signal ☐
- c. increases when the peak to peak load voltage increases ☐
- d. is maximum when the transistor is driven to cutoff ☒

?In a zener diode regulator, what value of load resistance results in the maximum zener current

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a. SHORT ($0\ \Omega$) ☐

b. $100\ \Omega$ ☐

c. 10K ☐

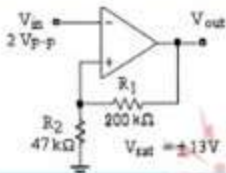
d. An Infinite resistance (open) ☒

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If the load resistance connected to a filtered power supply is decreased, what happens to the ripple voltage

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- a. If the load resistance connected to a filtered power supply is decreased, what happens to the ripple voltage ☐
- b. not affected ☒
- c. If the load resistance connected to a filtered power supply is increased, what happens to the ripple voltage ☐
- d. the same ☐



The output voltage with the inputs as shown

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a. $-V_{sat}$ ☐

b. 17.06 V_{p-p} ☐

c. $+V_{sat}$ ☒

d. 26 V_{p-p} ☐

In an op-amp integrator, the current through the input resistor flows into the:

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a. inverting input ☐

b. noninverting input ☐

c. bypass capacitor ☐

d. feedback capacitor ☒

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What causes the ripple voltage on the output of a capacitor-input filter

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a. The ripple voltage is caused by the slight discharging of the capacitor through the load resistor ☐

b. none ☐

c. The ripple voltage is caused by the slight charging and discharging of the capacitor through the load resistor ☐

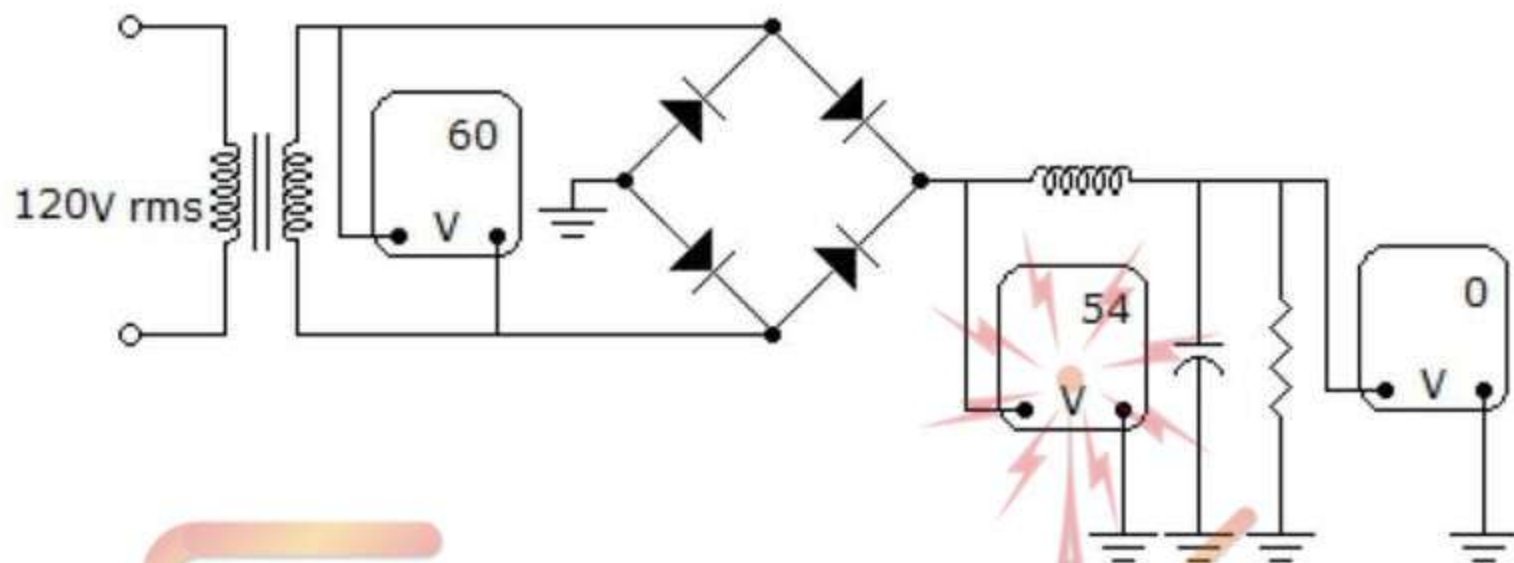
d. The ripple voltage is caused by the slight charging of the capacitor through the load resistor ☒

When operating the transistor, a dc biasing is necessary to establish the proper region of operation for ac amplification.

Select one:

- ☒ a. True
- ☐ b. False

52. Refer to the figure given below. If the voltmeter across the transformer reads 0 V, the probable trouble, if any, would be



- A. one of the diodes is open.
- B. a diode is shorted.
- C. an open transformer secondary.
- D. the filter capacitor is shorted.
- E. no trouble exists.

Answer: Option C

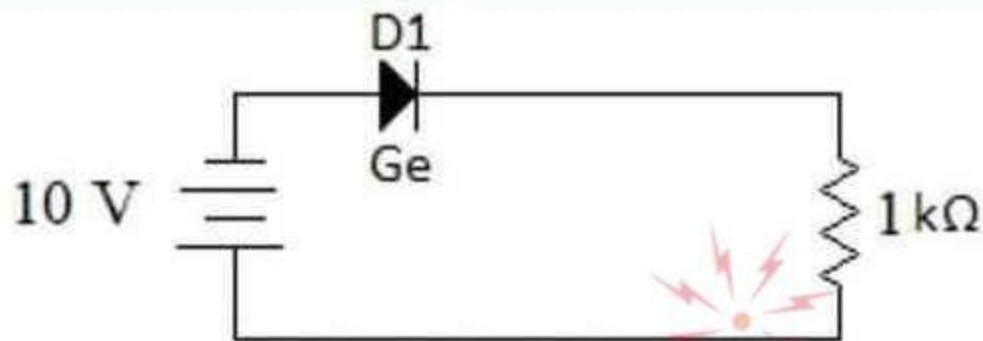
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For the positive clamper circuit and the input signal is square wave ($+4\text{ V}$ to -8 V) Which of the following is true for the output?

Select one

- ☐ a. The output waveform is a square wave with $+4\text{ V}$ level clamped to 0 V
- ☒ b. The output signal is a square wave with lowest level clamped to $+4\text{ V}$
- ☐ c. The output waveform is a V triangle with highest level clamped to 4 V
- ☐ d. The output waveform is a triangle wave with the lowest level clamped to 0 V

The value of VR is:

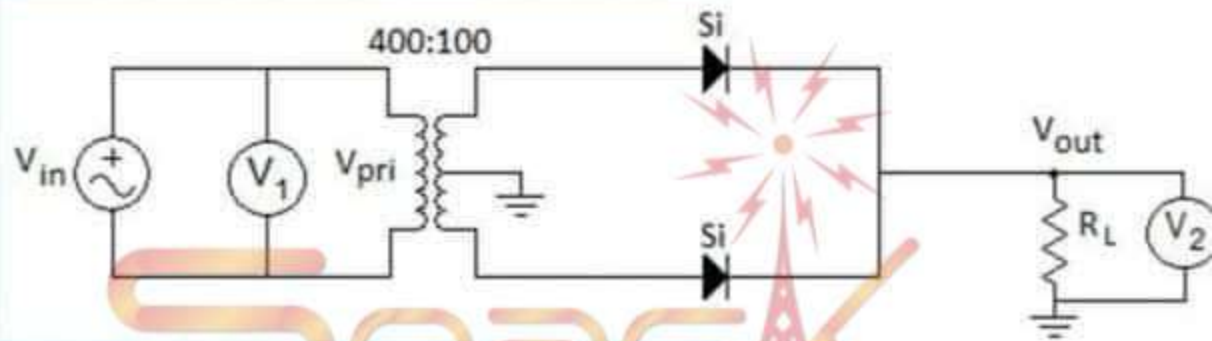


Select one:

- ☐ a. 10 V
- ☐ b. 9.3 V
- ☒ c. 0 V
- ☐ d. 3.7 V

$V_p(\text{sec}) = 20\text{V}$

Calculate the value of $V_p(\text{out})$



Select one:

- ☐ a. 5 V
- ☐ b. 14.14V
- ☐ c. 28.28V
- ☒ d. 9.3 V

What type of diode circuit is used to add or restore a dc level to an electrical signal?

Select one:

- ☐ a. IC voltage regulator
- ☐ b. none of the mentioned
- ☐ c. clamper
- ☒ d. clipper or limiter

To setup Common Drain you need the following

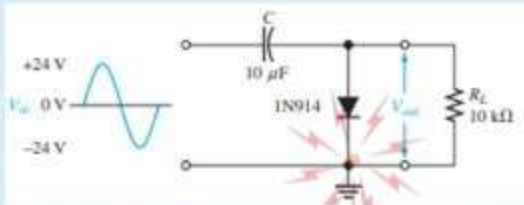
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a. Either Ac or DC ☒

b. Both DC and Ac ☐

c. Only Ac voltage ☐

d. Only DC voltage ☐



What is the output voltage that you would expect to observe across R_L .

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- a. 0.7–47.3 V ☐
- b. $+(24\text{ V} + 0.7\text{ V})$ ☐
- c. 0–48 v ☒
- d. $-(24\text{ V} - 0.7\text{ V})$ ☐

_____ A transistor can be checked using a(n)

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a. curve tracer ☐

b. ohmmeter ☐

c. digital meter ☐

d. Any of the above ☒

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How do you determine when a transistor is saturated? When a transistor is in cutoff .

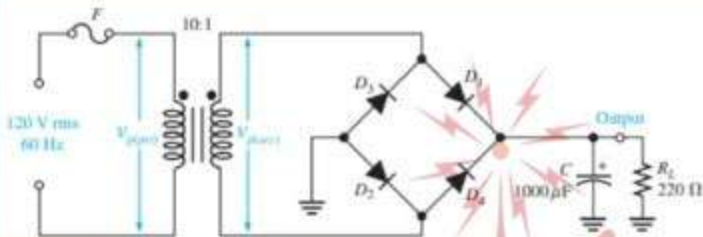
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a. $V_{CE\text{SAT}} = V_{CC}$... $V_{CE\text{CUT}} = 0\text{V}$ ☐

b. $V_{CE\text{SAT}} = V_{CC}$... $V_{CE\text{CUT}} = V_{CE\text{SAT}}$ ☒

c. $V_{CE\text{CUT}} = V_{CE\text{SAT}}$ ☐

d. $V_{CE\text{SAT}} = 0\text{V}$... $V_{CE\text{CUT}} = V_{CC}$ ☐

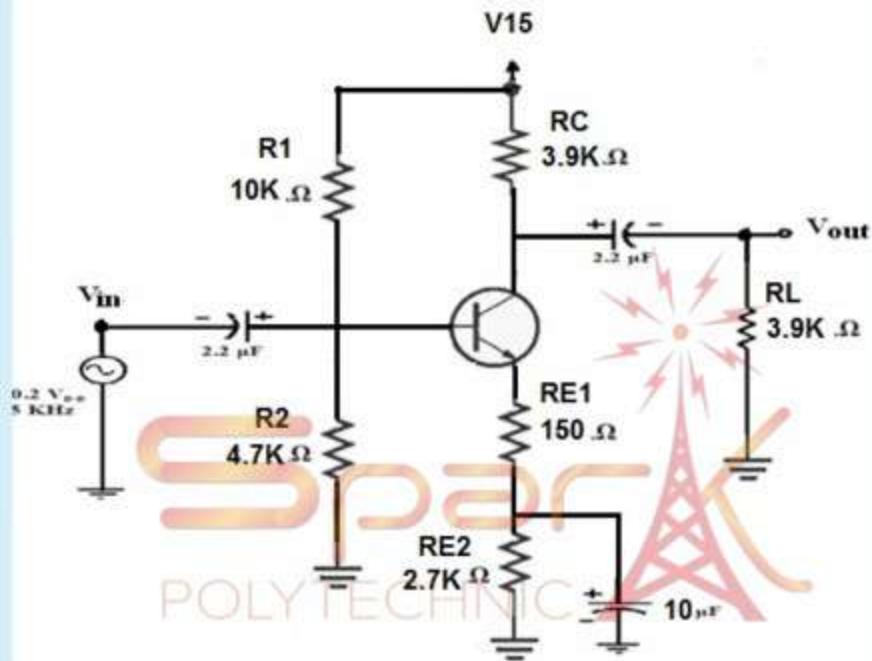


Determine the ripple factor for the filtered bridge rectifier with a load as indicated in Figure

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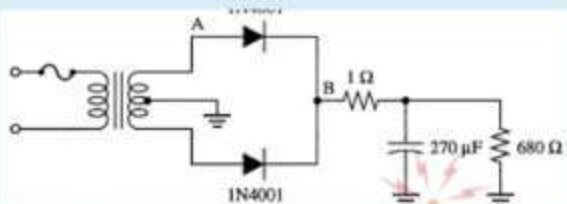
- a. 2.3 ☐
- b. 1.7 ☒
- c. 0.227 ☐
- d. 0.39 ☐

For the circuit shown find V_{CE}



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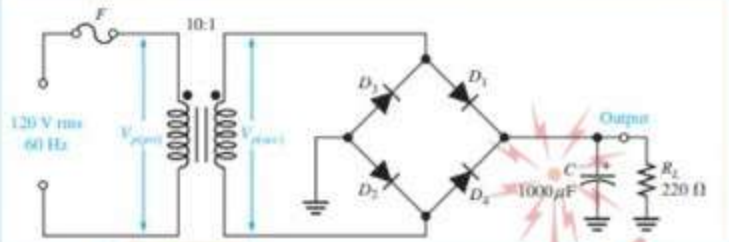
- a. 0.7 V ☒
- b. 5.28 V ☐
- c. 10.9 V ☐
- d. 9.38 V ☐



The voltage at point A with respect to ground is 12.7 V
 $= PIV$

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- a. 24.7 V ☐
- b. 12.7 V ☒
- c. 36.7V ☐
- d. 6.7 V ☐



The approximate dc value of the output voltage is

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a. 15.6 V ☐

b. 17 V ☐

c. 15.3 V ☒

d. 1.56V ☐

In a voltage-divider biased npn transistor circuit, you measure VCC at the collector and an emitter voltage 0.7 V less than the base voltage. Is the transistor functioning in cutoff, or is R_E open?

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- a. none ☐
- b. R_E is open because the BE junction of the transistor is still reverse -biased ☐
- c. R_E is open because the BE junction of the transistor is still forward-biased ☐
- d. R_E is short because the BE junction of the transistor is still forward-biased ☒

What is the maximum voltage across an unbiased positive silicon diode limiter during the positive alternation of the input voltage .



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- a. 5 V appears across the diode ☐
- b. 0.7 V appears across the diode ☒
- c. 0.5 V appears across the diode ☐
- d. PIV ☐

To limit the output voltage of a positive limiter to 5 V when a 10 V peak input is applied, , what value must the bias voltage be

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a. 0V ☐

b. The bias voltage must be 0.7 V ☐

c. The bias voltage must be $5\text{ V} + 0.7\text{ V} = 5.7\text{ V}$ ☐

d. The bias voltage must be $5\text{ V} - 0.7\text{ V} = 4.3\text{ V}$ ☒

What is the difference between input (line) regulation and load regulation?

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- a. Input regulation is a measure of the variation in output voltage over a range of output voltages. Load regulation is a measure of the variation in output voltage over a range of load current values. ☐
- b. Input regulation is a measure of the variation in output voltage over a range of input voltages. Load regulation is a measure of the variation in output voltage over a range of load current values. ☐
- c. Input regulation is a measure of the variation in input voltage over a range of input voltages. Load regulation is a measure of the variation in input voltage over a range of load current values. ☐
- d. all of above ☒

?At what point on the input cycle does the PIV occur .

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a. PIV across the diode occurs at the peak of the input when the diode is reversed biased ☐

b. ☐

$PIV = V_D$

c. none ☐

d. PIV across the diode occurs at the peak of the output when the diode is reversed biased ☒

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If the input to a comparator is a sine wave, the output is a:

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a. ramp voltage ☐

b. sawtooth wave ☐

c. sine wave ☒

d. rectangular wave ☐

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